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Physical Properties	Metric	English	Comments
Density	2.78 g/cc	0.100 lb/in³	AA; Typical

Mechanical Properties	Metric	English	Comments
Hardness, Brinell	125	125	500 kg load/10 mm ball
Hardness, Knoop	157	157	Estimated from Brinell
Hardness, Rockwell A	48	48	Estimated from Brinell
Hardness, Rockwell B	78	78	Estimated from Brinell
Hardness, Vickers	142	142	Estimated from Brinell
Tensile Strength, Ultimate	>= 415 MPa	>= 60200 psi	Wire, rod, and bar (rolled or cold finished); T62
	>= 425 MPa	>= 61600 psi	Wire, rod, and bar (rolled or cold finished)
	>= 427 MPa	>= 61900 psi	
	34.0 MPa @Temperature 371 °C	4930 psi @Temperature 700 °F	
	52.0 MPa @Temperature 316 °C	7540 psi @Temperature 601 °F	
	76.0 MPa @Temperature 260 °C	11000 psi @Temperature 500 °F	
	179 MPa @Temperature 204 °C	26000 psi @Temperature 399 °F	
	310 MPa @Temperature 149 °C	45000 psi @Temperature 300 °F	
	448 MPa @Temperature 100 °C	65000 psi @Temperature 212 °F	
	476 MPa @Temperature 24.0 °C	69000 psi @Temperature 75.2 °F	
	483 MPa @Temperature -28.0 °C	70100 psi @Temperature -18.4 °F	
	496 MPa @Temperature -80.0 °C	71900 psi @Temperature -112 °F	
	579 MPa @Temperature -196 °C	84000 psi @Temperature -321 °F	
	>= 435 MPa @Thickness 12.7 - 76.2 mm	>= 63100 psi @Thickness 0.500 - 3.00 in	Sheet and plate; T62
	>= 440 MPa @Thickness 0.254 - 12.7 mm	>= 63800 psi @Thickness 0.0100 - 0.499 in	Sheet and plate; T62
Tensile Strength, Yield	>= 315 MPa	>= 45700 psi	Wire, rod, and bar (rolled or cold finished); T62
	>= 345 MPa	>= 50000 psi	
	>= 345 MPa	>= 50000 psi	Wire, rod, and bar (rolled or cold finished)
	>= 345 MPa @Thickness 0.254 - 12.7 mm	>= 50000 psi @Thickness 0.0100 - 0.499 in	Sheet and plate; T62
	>= 345 MPa @Thickness 12.7 - 76.2 mm	>= 50000 psi @Thickness 0.500 - 3.00 in	Sheet and plate; T62
	28.0 MPa @Strain 0.2 % Temperature 371 °C	4060 psi @Strain 0.2 % Temperature 700 °F	
	41.0 MPa @Strain 0.2 % Temperature 316 °C	5950 psi @Strain 0.2 % Temperature 601 °F	
	62.0 MPa @Strain 0.2 % Temperature 260 °C	8990 psi @Strain 0.2 % Temperature 500 °F	
	131 MPa @Strain 0.2 % Temperature 204 °C	19000 psi @Strain 0.2 % Temperature 399 °F	
	248 MPa @Strain 0.2 % Temperature 149 °C	36000 psi @Strain 0.2 % Temperature 300 °F	
	372 MPa @Strain 0.2 % Temperature 100 °C	54000 psi @Strain 0.2 % Temperature 212 °F	
	393 MPa @Strain 0.2 % Temperature 24.0 °C	57000 psi @Strain 0.2 % Temperature 75.2 °F	
	400 MPa @Strain 0.2 % Temperature -28.0 °C	58000 psi @Strain 0.2 % Temperature -18.4 °F	
	407 MPa @Strain 0.2 % Temperature -80.0 °C	59000 psi @Strain 0.2 % Temperature -112 °F	
	469 MPa @Strain 0.2 % Temperature -196 °C	68000 psi @Strain 0.2 % Temperature -321 °F	
Elongation at Break	5.0 %	5.0 %	
	>= 5.0 %	>= 5.0 %	Wire, rod, and bar (rolled or cold finished)
	>= 5.0 %	>= 5.0 %	Wire, rod, and bar (rolled or cold finished); T62
	10 % @Temperature -80.0 °C	10 % @Temperature -112 °F	
	10 % @Temperature -28.0 °C	10 % @Temperature -18.4 °F	
	10 % @Temperature 24.0 °C	10 % @Temperature 75.2 °F	
	10 % @Temperature 100 °C	10 % @Temperature 212 °F	
	11 % @Temperature -196 °C	11 % @Temperature -321 °F	
	17 % @Temperature 149 °C	17 % @Temperature 300 °F	
	27 % @Temperature 204 °C	27 % @Temperature 399 °F	
	55 % @Temperature 260 °C	55 % @Temperature 500 °F	
	75 % @Temperature 316 °C	75 % @Temperature 601 °F	
	100 % @Temperature 371 °C	100 % @Temperature 700 °F	
	>= 5.0 % @Thickness 0.254 - 12.7 mm	>= 5.0 % @Thickness 0.0100 - 0.499 in	Sheet and plate; T62
	>= 5.0 % @Thickness 12.7 - 76.2 mm	>= 5.0 % @Thickness 0.500 - 3.00 in	Sheet and plate; T62
Modulus of Elasticity	72.4 GPa	10500 ksi	Estimated from other heat treatments
Poissons Ratio	0.33	0.33	Estimated from other heat treatments
Fatigue Strength	124 MPa @# of Cycles 5.00e+8	18000 psi @# of Cycles 5.00e+8	completely reversed; R. R. Moore Machine and specimen

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	>= 5.0 %	>= 5.0 %	Wire, rod, and bar (rolled or cold finished)
	>= 5.0 %	>= 5.0 %	Wire, rod, and bar (rolled or cold finished); T62
	10 % @Temperature -80.0 °C	10 % @Temperature -112 °F	
	10 % @Temperature -28.0 °C	10 % @Temperature -18.4 °F	
	10 % @Temperature 24.0 °C	10 % @Temperature 75.2 °F	
	10 % @Temperature 100 °C	10 % @Temperature 212 °F	
	11 % @Temperature -196 °C	11 % @Temperature -321 °F	
	17 % @Temperature 149 °C	17 % @Temperature 300 °F	
	27 % @Temperature 204 °C	27 % @Temperature 399 °F	
	55 % @Temperature 260 °C	55 % @Temperature 500 °F	
	75 % @Temperature 316 °C	75 % @Temperature 601 °F	
	100 % @Temperature 371 °C	100 % @Temperature 700 °F	
	>= 5.0 % @Thickness 0.254 - 12.7 mm	>= 5.0 % @Thickness 0.0100 - 0.499 in	Sheet and plate; T62
	>= 5.0 % @Thickness 12.7 - 76.2 mm	>= 5.0 % @Thickness 0.500 - 3.00 in	Sheet and plate; T62
Modulus of Elasticity	72.4 GPa	10500 ksi	Estimated from other heat treatments.
Poissons Ratio	0.33	0.33	Estimated from other heat treatments.
Fatigue Strength	124 MPa @# of Cycles 5.00e+8	18000 psi @# of Cycles 5.00e+8	completely reversed, R. R. Moore Machine and specimen.
Machinability	70 %	70 %	0-100 Scale (A=90; B=70; C=50; D=30; E=10)
Shear Modulus	27.0 GPa	3920 ksi	Estimated from similar Al alloys.
Shear Strength	283 MPa	41000 psi	

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.00000449 ohm-cm @Temperature 20.0 °C	0.00000449 ohm-cm @Temperature 68.0 °F	AA; Typical

Thermal Properties	Metric	English	Comments
CTE, linear	21.1 µm/m-°C @Temperature -50.0 - 20.0 °C	11.7 µin/in-°F @Temperature -58.0 - 68.0 °F	
	22.9 µm/m-°C @Temperature 20.0 - 100 °C	12.7 µin/in-°F @Temperature 68.0 - 212 °F	
	23.2 µm/m-°C @Temperature 20.0 - 100 °C	12.9 µin/in-°F @Temperature 68.0 - 212 °F	AA; Typical; average over range
	23.8 µm/m-°C @Temperature 20.0 - 200 °C	13.2 µin/in-°F @Temperature 68.0 - 392 °F	
	24.7 µm/m-°C @Temperature 20.0 - 300 °C	13.7 µin/in-°F @Temperature 68.0 - 572 °F	
Specific Heat Capacity	0.875 J/g-°C	0.209 BTU/lb-°F	
Thermal Conductivity	151 W/m-K	1050 BTU-in/hr-ft²-°F	AA; Typical at 77°F
Melting Point	502 - 638 °C	935 - 1180 °F	AA; Typical range based on typical composition for wrought products >= 1/4 in. thickness. Eutectic melting is not eliminated by homogenization.
Solidus	502 °C	935 °F	AA; Typical
Liquidus	638 °C	1180 °F	AA; Typical

Processing Properties	Metric	English	Comments
Annealing Temperature	413 °C	775 °F	
Solution Temperature	493 °C	919 °F	
Aging Temperature	191 °C	375 °F	8 to 16 hr at temperature

Component Elements Properties	Metric	English	Comments
Aluminum, Al	90.7 - 94.7 %	90.7 - 94.7 %	As remainder
Chromium, Cr	<= 0.10 %	<= 0.10 %	
Copper, Cu	3.8 - 4.9 %	3.8 - 4.9 %	
Iron, Fe	<= 0.50 %	<= 0.50 %	
Magnesium, Mg	1.2 - 1.8 %	1.2 - 1.8 %	
Manganese, Mn	0.30 - 0.90 %	0.30 - 0.90 %	
Other, each	<= 0.05 %	<= 0.05 %	
Other, total	<= 0.15 %	<= 0.15 %	
Silicon, Si	<= 0.50 %	<= 0.50 %	
Titanium, Ti	<= 0.15 %	<= 0.15 %	
Zinc, Zn	<= 0.25 %	<= 0.25 %	

References for this datasheet.

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